1. Chemistry of the Environment

1.1. Water

Chemical Tests for the Presence of Water

Test	Type of test	Positive result
Anhydrous Cobalt (II) Chloride	Chemical	It turns from blue to pink
Anhydrous Copper (II) Sulfate	Chemical	It turns from white to blue
Test Melting and Boiling Point	Physical	M.P at 0℃ and B.P at 100℃

- **Distilled Water is used** in practical chemistry rather than tap Water because it has fewer chemical impurities.

Water from Natural Sources

Water is an essential source in the natural world. However, with the changing world and massive urbanisation. **Our water may contain substances such as:**

1. Dissolved Oxygen (**this is important for aquatic life**)
2. Metal Compounds (Provide **essential minerals for life - however, some are toxic**)
3. Plastics (**harm aquatic life**)
4. Sewage (**contains harmful microbes which cause diseases**)
5. Harmful microbes.
6. Nitrates from fertilisers
7. Phosphate from fertilisers and detergents **(leads to deoxygenation of water and damage to aquatic life (Eutrophication)**

Treatment of Domestic Water Supply

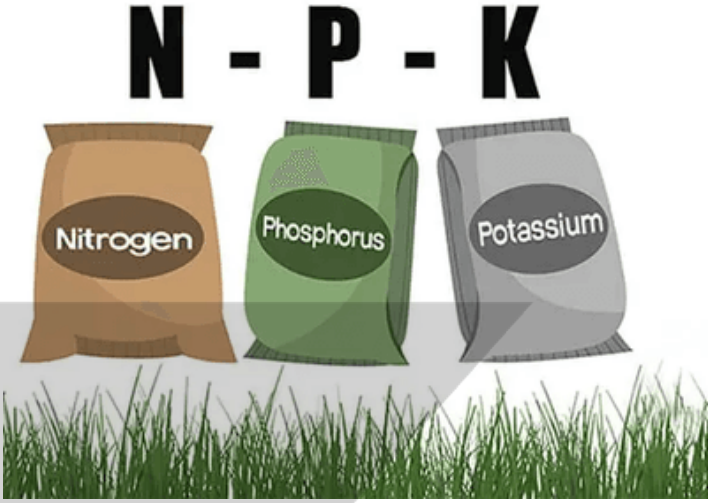
1. Water is pumped into screens to **remove solid, insoluble impurities.**
2. A **sedimentation** process makes small clay pieces stick together and are then removed.
3. The water then undergoes **filtration** through layers of sand and gravel to **remove larger, insoluble debris**.
4. **Carbon** is also added into filtered water to **remove taste and odour**.
5. The **chlorination process** adds chlorine gas bubbled into the water to **kill bacteria and other microbes**; the acidic effect on the water is reversed by adding an alkali, sodium hydroxide.

Step one and five, chlorine is added before going to domestic areas.

1.2. Fertilisers

Fertilisers: Substances added to the soil and taken up by plants to increase crop yield.

Fertilisers contain substances such as **nitrates** and **ammonium salts**.



N.P.K - Nitrogen, Phosphorus, and Potassium Fertilisers are found inside fertilisers, essential to **improve plant growth.**

Functions of Elements

1. Nitrogen - Makes chlorophyll and protein. Promotes healthy leaves
2. Phosphorus - Promotes healthy roots
3. Potassium - Promotes growth and healthy fruits and flowers

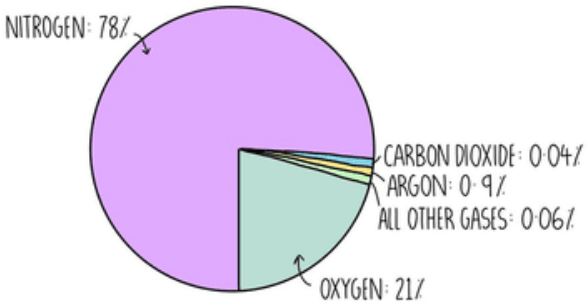
Reaction with any alkali substance (except ammonia) displaces ammonia from its compound, for example:

Calcium hydroxide + Ammonium chloride → Calcium chloride + Ammonia + Water

1.3. Air Quality and Climate

The pie chart below presents the components present in clean air:

- Primary: **Nitrogen (78%), Oxygen (21%)**
- Secondary: **Noble gases (mainly Argon) and Carbon Dioxide (1%)**



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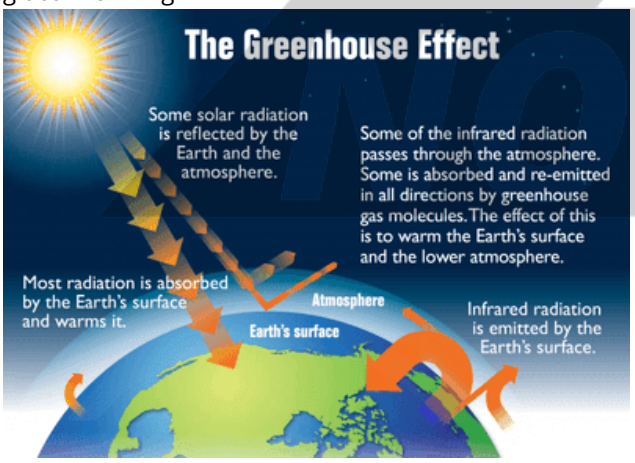
Air Pollutants and their Adverse Effects

Pollutant	Source	Negative impact
Carbon monoxide (CO)	Incomplete combustion of carbon-containing fuels (ex. Internal combustion engines)	Binds with haemoglobin, constricting oxygen supply in cells; leads to fatigue/ death
Carbon Dioxide (CO_2)	Complete Combustion of Carbon Containing Fuels	Increased global warming leads to climate change.
Methane (CH_4)	Decomposition of vegetation and waste gases from digestion in animals	Increased global warming leads to climate change.
Sulfur Dioxide (SO_2)	Combustion of fossil fuels which contain sulfur compounds	It causes acid rain.
Nitrogen Oxides (NO_2)	High temperatures that trigger a reaction between N_2 and O_2 (from air)	Causes respiratory problems and photochemical smog; contributes to acid rain
Lead Compounds	Combustion of leaded fuels	Damages brain and nerve cells in young children

Impact of Greenhouse Gases

1. **Short wavelength radiation** from the Sun reaches the Earth's surface
2. Some thermal energy is **absorbed and heats oceans/lands**
3. Earth radiates some thermal energy **as more prolonged wavelength radiation**
4. Greenhouse gases absorb some of the infrared radiation and **re-emit in all directions**
5. Some infrared radiation comes back to Earth's surface, and this reduces the heat loss to space and leads to global warming



Strategies to Reduce the Effect of Environmental Issues

1. **Climate Change**: planting trees, reducing livestock farming, decreasing the use of fossil fuels, increasing the use of hydrogen and renewable energy, e.g. wind, solar
2. **Acid Rain**: use of catalytic converters in vehicles, reducing emissions of sulfur dioxide by using low-sulfur fuels and flue gas desulfurisation with calcium oxide

Photosynthesis

Photosynthesis: the reaction between carbon dioxide and water to produce glucose and oxygen in the presence of chlorophyll and using energy from light.

Word Equation: **Carbon Dioxide + Water → Glucose + Oxygen**

Balanced Chemical Equation:

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

1.4. Catalytic Converters

1. Present in car exhausts; contains transition metal catalysts of platinum and rhodium
2. Aids redox reactions to neutralize toxic pollutants formed as a result of incomplete fuel combustion: (a) Carbon Monoxide, (b) Nitrogen Oxides, (c) Unburned hydrocarbons
3. Reaction equations:

   (a) $2CO + O_2 \rightarrow 2CO_2$

   (b) $2NO + 2CO \rightarrow N_2 + 2CO_2$

   (c) $C_8H_{18} + 12\frac{1}{2}O_2 \rightarrow 8CO_2 + 9H_2O$

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